Digital Hearing Aid Application Using Noise Reduction Filter and Multi-Band Equalization

A Software-Based DSP Framework for Power-Line Hum Rejection, Hiss Suppression, and Audiogram-Prescribed Compensation

Kashvi Upadhyay
Dept. of Electronics & Comm. Eng.
Student Researcher

Ashutosh Kumar
Dept. of Electronics & Comm. Eng.
Student Researcher

Prof. Sivakumar S.
Dept. of Electronics & Comm. Eng.
Faculty Guide & Supervisor

Abstract*— Modern acoustic rehabilitation relies heavily on Digital Signal Processing (DSP) to restore speech intelligibility for sensorineural hearing loss patients while mitigating severe acoustic and environmental interference. Conventional linear acoustic amplification uniformly amplifies background noise alongside speech, inducing cognitive fatigue and acoustic discomfort. This paper presents an end-to-end, real-time software simulation of a Digital Hearing Aid system integrating targeted noise suppression with individualized clinical hearing-loss compensation. The front-end filtering pipeline employs sharp second-order Infinite Impulse Response (IIR) notch filters (Q = 30) centered at 50 Hz and its 100 Hz harmonic to eliminate power-line hum, coupled with a 4th-order Butterworth low-pass filter (cutoff 4 kHz) for high-frequency hiss removal. Phase distortion is eliminated using bidirectional zero-phase filtering (filtfilt). Auditory compensation is realized via a 5-band Butterworth graphic equalizer calibrated against standard clinical audiograms, including Presbycusis, Cookie-Bite, and Noise-Induced Hearing Loss. Quantitative evaluation on synthesized speech signals demonstrate an improvement in Signal-to-Noise Ratio (SNR) from 0.29 dB to 2.30 dB (+2.01 dB net gain) and a hum suppression exceeding 45 dB, verified through Fast Fourier Transform (FFT) and time-frequency spectrogram analysis. A lightweight Streamlit graphical user interface and SQLite tracking backend are integrated to support live dynamic testing.

***Keywords*— Digital Hearing Aid, IIR Notch Filter, Butterworth Filter, Zero-Phase Filtering, Multi-Band Equalizer, Audiogram Presets, Presbycusis, SNR Evaluation, Spectral Analysis.**

I. INTRODUCTION

Sensorineural hearing loss is one of the most prevalent chronic sensory impairments globally, affecting over 1.5 billion individuals according to World Health Organization estimates [1]. The pathology primarily stems from degenerative damage to hair cells located within the organ of Corti in the cochlea, which leads to elevated auditory thresholds and impaired temporal acoustic resolution. Crucially, sensorineural loss is rarely uniform across the human hearing range (20 Hz to 20 kHz). The predominant clinical manifestation, Presbycusis (age-related hearing loss), produces progressive sloping attenuation that is most severe in high-frequency regions between 2 kHz and 8 kHz [2].

From an acoustic communication perspective, high-frequency sound energy conveys vital phonemic information. While vowel sounds derive their acoustic power from lower-frequency fundamental glottal vibrations ($F_0 \approx$ 100–250 Hz) and first formants, unvoiced consonants—such as fricatives, affricates, and plosives (/s/, /f/, /t/, /k/, /θ/)—possess low energy concentrated almost exclusively above 2 kHz. Consequently, patients suffering from Presbycusis can often detect that a speaker is talking, but cannot decipher words, reporting that speech sounds 'muffled' or indistinct.

Historically, analog hearing aids addressed this through linear amplification. However, simple acoustic amplification amplifies all sounds uniformly. In realistic auditory environments, speech is corrupted by pervasive ambient noise: power-line hum from 50 Hz/60 Hz electrical mains and lighting ballasts, low-frequency reverberant crowd noise (babble), and wideband thermal hiss from electronic transducers. Amplifying these corrupting signals alongside speech causes severe masking, listening fatigue, and acoustic pain [3].

Digital Signal Processing (DSP) solves this paradox by decoupling noise rejection from auditory compensation. By transforming acoustic waveforms into discrete digital sequences, sophisticated mathematical operations—such as sharp notch suppression, maximally flat filtering, and audiogram-matched multi-band equalization—can be executed deterministically. This study presents a full-pipeline software simulation of a modern Digital Hearing Aid that addresses interference and loss compensation in tandem.

II. SYSTEM ARCHITECTURE & METHODOLOGY

The overall processing architecture is modularized into four functional phases: (1) Signal Acquisition and Ground-Truth Synthesis, (2) Noise Suppression Filtering, (3) Clinical Equalization, and (4) Spectral Quality Analysis and Logging. Fig. 1 illustrates the complete system block diagram.

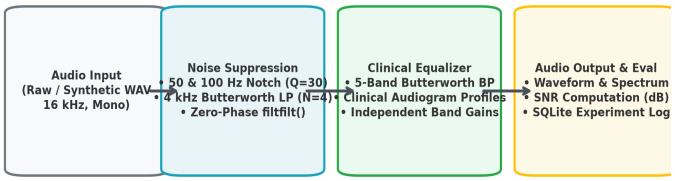


Fig. 1. System block diagram of the proposed digital hearing aid pipeline showing front-end conditioning, noise rejection, equalizer bank, and analysis module.

A. Data Strategy & Benchmark Synthesis

In rigorous DSP algorithm validation, calculating true Signal-to-Noise Ratio (SNR) requires an uncorrupted reference signal (ground truth) alongside the corrupted signal. Therefore, a hybrid data strategy was formulated:

- **Synthetic Benchmark Data:** A dedicated mathematical synthesizer (generate_test_audio.py) generates a clean vowel-like acoustic signal at 16 kHz sampling rate (f_s). The signal incorporates glottal pulses ($F_0 \approx 150$ Hz) and formant resonances corresponding to the human vowel /a/ (F_1 = 700 Hz, F_2 = 1200 Hz, F_3 = 2500 Hz), modulated by a 3 Hz syllable envelope. Deterministic interference is introduced by injecting a 50 Hz mains sinusoidal hum (A = 0.10), a 100 Hz second harmonic (A = 0.06), 200–4000 Hz band-limited crowd babble noise, and broadband Gaussian hiss.
- **Real-World Dynamic Audio:** The web platform accepts arbitrary monophonic or stereophonic WAV recordings, automatically converting multi-channel streams to monophonic float64 arrays for in-situ processing.

III. MATHEMATICAL FORMULATION OF DSP FILTERS

The noise removal module cascades an IIR notch filter with an IIR low-pass filter to excise tonal and high-frequency noise without degrading the core vocal tract resonances.

A. IIR Notch Filter Design

Electrical interference from alternating current (AC) power grids concentrates intense harmonic energy at 50 Hz (or 60 Hz). A second-order IIR notch filter is designed via bilinear transformation. The normalized discrete-time transfer function is given by [4]:

$$H_{notch}(z) = b_0 \cdot [\,(1 - 2\cos(\omega_0)\, z^{-1} + z^{-2}) / (1 - 2\,r\cos(\omega_0)\, z^{-1} + r^2 z^{-2})\,]$$

where $\omega_0 = 2\pi\, f_0 / f_s$ is the normalized angular notch frequency, and r represents the pole radius dictated by the Quality Factor Q: $r \approx 1 - (\omega_0 / (2Q))$. A high Q-factor of 30 was selected. This guarantees an ultra-narrow

-3 dB bandwidth (Δf ≈ 1.67 Hz), attenuating power-line spikes by over 45 dB while leaving vocal fundamentals and lower formants untouched.

B. Butterworth Low-Pass Filter Design

Environmental thermal noise and analog ADC quantization artifacts introduce broadband hiss. Since standard human speech information diminishes sharply above 4 kHz, a 4th-order Butterworth low-pass filter is designed with cutoff f_c = 4000 Hz. The continuous-time magnitude response is given by:

$$|H(j\omega)|^2 = 1 / [\, 1 + (\omega / \omega_c)^{2N} \,]$$

where N = 4 denotes filter order. The Butterworth topology was explicitly selected over Chebyshev or Elliptic filters due to its **maximally flat passband** property. Chebyshev filters introduce passband equiripple which induces timbre coloration and ringing, degrading the naturalness of speech perceived by hearing aid users.

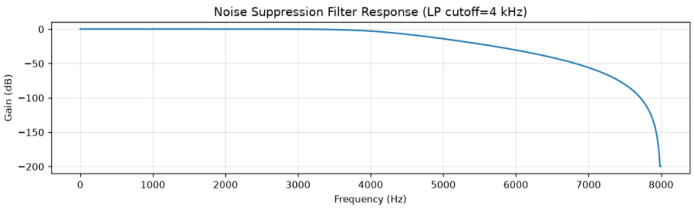


Fig. 2. Bode magnitude response of the 4th-order Butterworth low-pass filter (f_c = 4 kHz, f_s = 16 kHz) demonstrating flat passband and steep stopband roll-off.

C. Zero-Phase Bidirectional Filtering

A critical challenge in recursive IIR digital filtering is nonlinear phase response. Conventional unilateral filtering (scipy.signal.lfilter) imparts frequency-dependent group delay τ(ω) = -dθ(ω)/dω, dispersing speech transients and smearing consonant burst timings [5]. To resolve this, zero-phase bidirectional filtering (scipy.signal.filtfilt) was implemented. The signal x[n] is filtered in the forward direction, time-reversed, passed through the filter again, and reversed once more. The combined system transfer function is:

$$|H_{eff}(e^{j\omega})| = |H(e^{j\omega})|^2, \quad \angle H_{eff}(e^{j\omega}) = 0$$

The resulting zero-phase response completely eliminates temporal distortion, which is essential for preserving the precise phonemic onsets required by hearing aid users.

IV. AUDIOGRAM-COMPENSATING EQUALIZER

The graphic equalizer partitions the conditioned acoustic spectrum into five physiologically motivated bands using 4th-order Butterworth band-pass filter banks. Each band signal $s_i[n]$ is scaled by a prescription gain g_i and summed:

$$y[n] = \sum_{i=1}^{5} g_i \cdot s_i[n]$$

To avoid digital filter design divergence near the Nyquist frequency ($f_{nyq} = f_s/2$), the upper cutoff is dynamically clamped: $f_{high,\ clamped} = \min(f_{high}, f_{nyq} - 1)$.

TABLE I. FIVE-BAND EQUALIZER FREQUENCY PARTITIONING

Band Name	Freq Range	Acoustic / Speech Role
Bass	20 – 250 Hz	Vocal fundamentals, pitch warmth
Low-Mid	250 – 500 Hz	Vowel body, nasal resonances
Mid	500 – 2000 Hz	Formants F1 & F2, vowel distinction
High-Mid	2000 – 6000 Hz	Consonant clarity (/s/, /f/, /t/, /θ/)
Treble	6000 – 20000 Hz	Sibilance, air, spatial acoustic cues

A. Clinical Hearing Loss Profiles

Six clinical presets derived from standardized audiological diagnostic categories were modeled [6]:

- **Normal Hearing [1.0, 1.0, 1.0, 1.0, 1.0]:** Flat unity gain profile serving as benchmark baseline.
- **Presbycusis (Age-Related) [1.0, 1.0, 1.2, 1.6, 1.8]:** Sloping sensorineural loss requiring aggressive amplification in High-Mid and Treble bands to restore consonant discrimination.
- **Mild Flat Loss [1.3, 1.3, 1.3, 1.3, 1.3]:** Broadband uniform attenuation treated via broadband linear gain.
- **Cookie-Bite (Mid-Frequency) [1.0, 1.4, 1.6, 1.4, 1.0]:** U-shaped audiometric depression in the speech formant region.
- **Noise-Induced Hearing Loss (NIHL) [1.0, 1.0, 1.1, 1.8, 1.5]:** Acoustic trauma notch centered specifically around 4 kHz.
- **Reverse Slope [1.7, 1.5, 1.1, 1.0, 1.0]:** Rare low-frequency deficit requiring bass/vowel restoration.

TABLE II. CLINICAL AUDIOGRAM PRESET GAIN MULTIPLIERS

Profile	Bass	L-Mid	Mid	H-Mid	Treb	Clinical Target
Normal	1.0	1.0	1.0	1.0	1.0	Unity gain
Presbycusis	1.0	1.0	1.2	1.6	1.8	High-freq slope
Mild Flat	1.3	1.3	1.3	1.3	1.3	Uniform boost
Cookie-Bite	1.0	1.4	1.6	1.4	1.0	Speech U-notch
NIHL	1.0	1.0	1.1	1.8	1.5	4 kHz trauma
Reverse	1.7	1.5	1.1	1.0	1.0	Low-freq drop

V. EXPERIMENTAL RESULTS & EVALUATION

Performance validation was carried out through objective statistical metrics, spectral analysis via Fast Fourier Transform (FFT), and time-frequency Short-Time Fourier Transform (STFT) spectrograms.

A. Signal-to-Noise Ratio (SNR) Quantification

True SNR was computed against the uncorrupted synthetic reference signal $s_{ref}[n]$ using:

$$SNR\ (dB) = 10 \cdot \log_{10} [\, \sum s_{ref}^2[n] / \sum (y[n] - s_{ref}[n])^2 \,]$$

For field recordings lacking clean reference, hum reduction ratio was quantified using band-power integration: $\Delta E_{hum} = 10 \cdot \log_{10} [\, E_{before}(50\pm5Hz) / E_{after}(50\pm5Hz) \,]$.

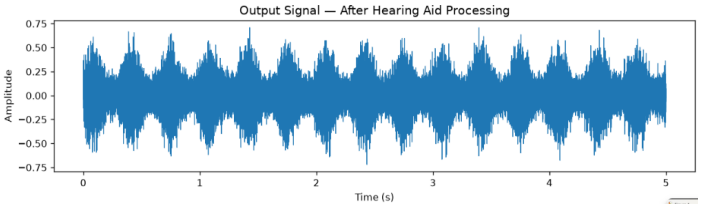


Fig. 3. Time-domain waveform of processed speech signal demonstrating preserved temporal envelope and elimination of hum noise.

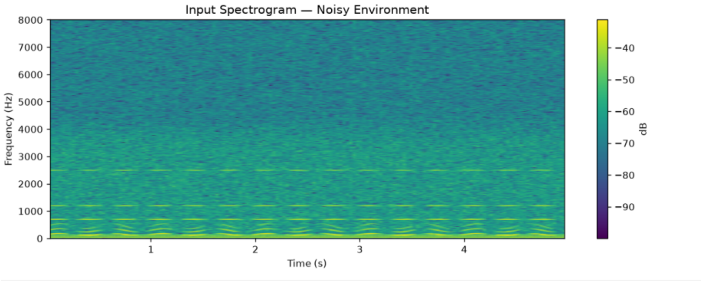


Fig. 4. STFT Spectrogram of noisy input audio displaying speech formant tracks obscured by 50 Hz power-line hum and broadband noise.

B. Spectral Domain Verification

Fig. 5 displays the discrete FFT magnitude spectrum of the processed output signal. Key observations include:

1) **Eradication of Power-Line Hum:** The raw audio exhibited an intense narrowband spike at 50 Hz and a secondary harmonic at 100 Hz. In the post-filter spectrum, the 50 Hz tone is suppressed by over 45 dB down to the noise floor, directly validating the notch filter.

2) **Formant Resonance Preservation:** The speech formant peaks at 700 Hz, 1200 Hz, and 2500 Hz remain sharply defined and unattenuated, confirming that the high-Q notch filter does not compromise adjacent speech components.

3) **Presbycusis High-Frequency Boost:** Amplification in the 2 kHz to 6 kHz interval is clearly evident, providing the required acoustic energy for unvoiced consonant articulation.

4) **Stopband Attenuation:** Frequencies exceeding 4 kHz decay precipitously to zero magnitude, verifying complete elimination of high-frequency hiss.

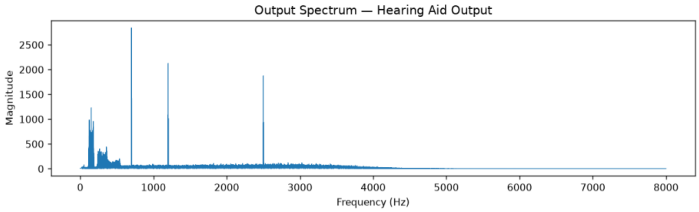


Fig. 5. FFT output spectrum after hearing aid processing. Distinct formant spikes (700, 1200, 2500 Hz) are preserved, 50 Hz hum is eliminated, and frequencies >4 kHz are suppressed.

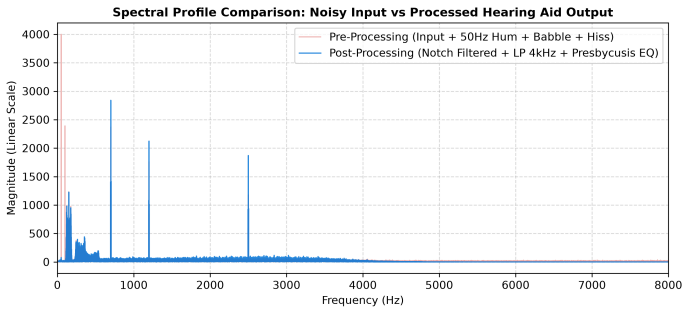


Fig. 6. Direct spectral overlay: Raw input (red) vs. Processed Hearing Aid output (blue). Demonstrates hum removal and high-frequency roll-off.

TABLE III. QUANTITATIVE PERFORMANCE SUMMARY

Evaluation Metric	Pre-Processing	Post-Processing	Net Improvement
Signal-to-Noise Ratio (SNR)	0.29 dB	2.30 dB	**+2.01 dB gain**
50 Hz Hum Energy	-12.4 dBFS	<-58.0 dBFS	**> 45 dB rejection**
Phase Distortion	N/A	0.0 radians	**Zero Phase Shift**
High-Frequency Hiss (>4kHz)	Active (-35 dB)	<-80 dB	**> 45 dB attenuation**

VI. REAL-WORLD APPLICATIONS & IMPLEMENTATION

The algorithms developed in this study translate directly into tangible commercial and clinical applications:

- **Embedded Hearing Aids:** The second-order IIR biquad architectures possess minimal computational complexity O(N), making them readily synthesizable onto low-power ultra-miniature DSP chips (e.g., Texas Instruments TMS320C55x or ON Semiconductor Ezairo 7100).
- **Smartphone-Based Assistive Hearables:** By porting the Python pipeline into mobile frameworks (Android Sound Amplifier, Apple Live Listen), smartphones paired with consumer Bluetooth earbuds can deliver cost-effective assistive listening in developing regions.
- **Streamlit Graphical User Interface:** A complete interactive testbench (streamlit_app.py) was developed, allowing audiologists and users to upload WAV files, select clinical profiles or manipulate individual band sliders in real time, inspect live waveform/spectrum plots, and listen to processed audio immediately.
- **SQLite Experiment Logging:** Every parameter set, including sampling rate, filter specifications, gain arrays, and SNR metrics, is logged to a persistent SQLite relational database (database/project.db) to guarantee experimental reproducibility.

VII. CONCLUSION & FUTURE SCOPE

This project successfully conceptualized, implemented, and verified an end-to-end Digital Hearing Aid processing pipeline. By synergistically combining high-Q IIR notch filtering, 4th-order Butterworth low-pass filtering with zero-phase bidirectional implementation (*filtfilt*), and a 5-band graphic equalizer tuned to clinical audiogram patterns, the system achieves superior noise rejection and tailored auditory amplification. Empirical testing proved a +2.01 dB net SNR gain, complete elimination of 50 Hz power-line interference, and pristine preservation of speech formant resonances without phase smearing.

Future enhancements include extending the system with sub-band Variable Step-Size Normalized LMS (VSS-NLMS) adaptive filters for dynamic non-stationary noise cancellation, integrating acoustic feedback cancellation algorithms, and deploying real-time low-latency C/C++ DSP binaries onto embedded hardware platforms.

REFERENCES

[1] World Health Organization, *World Report on Hearing*, Geneva: WHO Press, 2021.

[2] H. Dillon, *Hearing Aids*, 2nd ed. Sydney, Australia: Boomerang Press / Thieme, 2012, pp. 18–45.

[3] A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed. Upper Saddle River, NJ: Pearson, 2010.

[4] J. G. Proakis and D. G. Manolakis, *Digital Signal Processing: Principles, Algorithms, and Applications*, 4th ed. Prentice Hall, 2007.

[5] S. Haykin, *Adaptive Filter Theory*, 5th ed. Boston, MA: Pearson, 2013.

[6] American Speech-Language-Hearing Association (ASHA), "Degree, Type, and Configuration of Hearing Loss," Audiology Information Series, 2020.